Vehicle Rental Management System

Team 1: Mobile Application Development Lab

III MCA A – A2

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Introduction

The Vehicle Rental Management System is planned to be an Android-based mobile application developed in Android Studio using Java. The application will be designed to simplify the process of renting vehicles by providing a user-friendly platform for customers to browse the available vehicles and make bookings. Administrators can view and manage vehicle inventory, customer information, and booking records.

Problem Statement

Traditional offline vehicle rental businesses manually write vehicle records down on paper and manage bookings via phone calls or physical visits. These methods are time-consuming, can have human made errors, and are difficult to manage. Customers also face inconvenience in checking if vehicles are available and in booking vehicles. The Vehicle Rental Management System replaces the traditional manual rental process with an automated software-based solution, reducing manual paperwork, reducing errors, and improving efficiency for both customers and vehicle rental business owners.

Objectives

- To develop an Android-based mobile application using Android Studio and Java for vehicle rental management.
- To develop a user-friendly interface for customers.
- To allow customers to register, log in, search vehicles, book vehicles, cancel bookings, and view the prices for renting.
- To allow administrators to add, update, remove vehicle information, and to view customer bookings and payments.
- To maintain rental booking and customer records.
- To improve operational efficiency.
- To make the entire process easier for customers and vehicle rental business owners.

Software and Hardware Requirements

- Software Requirements:
  - Android Studio (2026.1.2- Quail 2)
  - Android SDK
  - Java (22.0.2)
  - MySQL Database
- Hardware Requirements:
  - Android Mobile (Android 7.0)
  - Computer/Laptop
  - Internet Connectivity

Modules

- User Module:
  User registration
  User login
  Search vehicles
  View prices
  Book vehicle
  Cancel booking
  View booking history

Update profile
Write feedback

- Admin Module:
  Admin login
  Add new vehicles
  Edit vehicle details
  Delete vehicles
  View and Delete customer accounts
  View customer bookings
  View customer payments
  View customer feedback

Conclusion

The Vehicle Rental Management System is a practical Android application that digitizes the vehicle rental process and makes the entire vehicle rental process easier for customers and business owners.